

Branch Specialization Checkpoint

Toy Competition Weight Sweep

Branch Specialization Project

2026-05-22

Checkpoint reason: the weight sweep gives a sharper mechanism for Phase 3.

Phase 3 RQ: Does structural heterogeneity cause stable functional specialization?

The layout-permutation result showed that the local role follows the 64-dim head, while induction is layout-sensitive.

This checkpoint asks:

If local-task pressure is reduced, does induction reclaim the 64-dim slot?

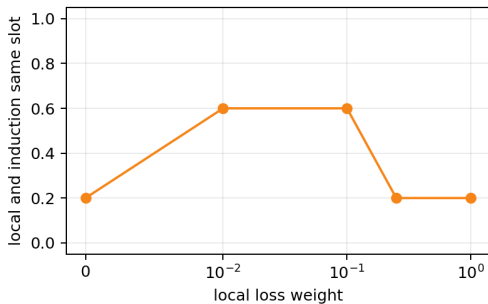
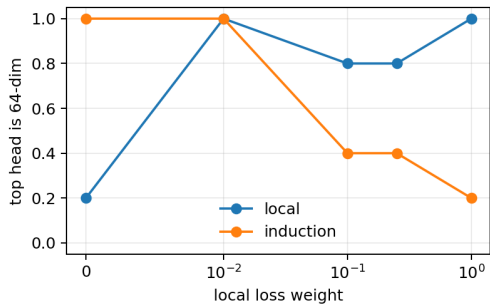
This directly tests capacity-slot competition.

Experiment Setup

- Layout: `hetero4_64second` = [16, 64, 16, 32].
- This was diagnostic because equal weights gave induction-on-64 in only 1/5 seeds.
- Induction weight fixed at 1.0.
- Local weight swept: 0.00, 0.01, 0.10, 0.25, 1.00.
- Five seeds per condition, 1200 training steps per seed.
- Role score is single-head causal ablation loss delta.

Main Evidence

hetero4_64second: role slots under local-weight sweep



Summary Table

Local weight	Local acc.	Induction acc.	Local top 64	Induction top 64	Same slot
0.00	0.2264	0.9981	0.20	1.00	0.20
0.01	0.9963	0.9976	1.00	1.00	0.60
0.10	0.9998	0.9976	0.80	0.40	0.60
0.25	0.9999	0.9980	0.80	0.40	0.20
1.00	1.0000	1.0000	1.00	0.20	0.20

Induction uses the 64-dim slot when local pressure is absent or tiny, but is often displaced once local pressure becomes substantial.

What Changed

Supported:

- The [16, 64, 16, 32] layout can put induction on the 64-dim head.
- Local pressure changes role allocation among structural slots.
- Heterogeneous dimensions create capacity attractors, not fixed semantics.

Nonlinear detail:

- At local weight 0.01, local learns and both roles use 64-dim heads in 5/5 seeds.
- At local weights 0.10, 0.25, and 1.00, induction often moves to 16-dim or 32-dim slots.

Best current mechanism:

Structural heterogeneity creates high-capacity attractor slots; weak task pressure can cohabit those slots, while stronger competing pressure can displace one role into secondary slots.

This strengthens the architectural-intervention story, but narrows the claim. The result is about symmetry breaking, capacity attraction, and task competition, not a fixed dimension-to-function semantic taxonomy.

Continue Phase 3 with the capacity-slot competition framing.

Next decisive tests:

- 1 Repeat the weight sweep across all four placements of the 64-dim head.
- 2 Add a matched-total-dimension condition with two medium-large heads.
- 3 Test whether role displacement weakens when there is more than one attractive slot.

If those hold, the causal story becomes: heterogeneity breaks permutation symmetry, creates capacity attractors, and task pressure allocates roles.

Sources and Provenance

- Report: `doc/phase3_toy_competition_weight_sweep.md`
- Training script: `scripts/toy_competition_head_dim_intervention.py`
- Analysis script: `scripts/analyze_competition_weight_sweep.py`
- Run outputs: `results/phase3_toy_competition_weight_lw*/`
- Combined analysis: `results/phase3_toy_competition_weight_sweep/`
- Deck data snapshot: `presentations/2026-05-22-0921-weight-sweep/data/`